# Provenance, Not Behaviour: A Serialisation Artifact in Edge-IIoTset and a Leakage-Free Benchmark for Precision-Agriculture Intrusion Detection

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**Abstract**

Edge-IIoTset has become the reference benchmark for machine-learning intrusion detection in the industrial Internet of Things, and the literature built on it reports accuracies clustered above 99%. We show that a large share of that performance is not intrusion detection. The preprocessing recipe distributed with the dataset instructs researchers to one-hot encode seven categorical columns. Four of those columns separate attack from normal traffic with an accuracy of 1.0000 on their own, through nothing more than the spelling of the placeholder written for an absent protocol field: the string `0` in the normal-traffic branch of the dataset build against the string `0.0` in the attack branch. The label is therefore recoverable from a serialisation artifact that encodes file provenance, without any network behaviour being modelled. The artifact is present in both curated subsets, separating every one of 157,800 rows in the ML subset and every one of 2,219,201 rows in the DNN subset, so both the classical-ML and the deep-learning branches of this literature are exposed to it. Under 5-fold $\times$ 3-repeat stratified cross-validation, five of six standard classifiers attain exactly $1.0000 \pm 0.0000$ accuracy across all fifteen folds and the sixth attains 0.99998, including Gaussian naive Bayes and logistic regression—models far too weak to genuinely solve the task. A 1D-CNN and a deep MLP of the kind that produce the literature's headline results reach 1.0000 accuracy on a single stratified split under the same recipe. Under a corrected protocol that canonicalises placeholder spellings, naive Bayes falls by 0.3005 macro-F1 and the strongest model settles at $0.9503 \pm 0.0011$. The leak is not a property of one-hot encoding: label, ordinal and frequency encoding all yield the same 1.0000

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single-column accuracy.

We then address a second obstacle to domain-specific work: the curated subsets cannot support an agricultural study at all, because Modbus is absent from them (0 of 157,800 rows in the ML subset; 150 of 2,219,201 in the DNN subset) and per-device identity has been stripped to a single surviving MQTT topic. We rebuild the benchmark from the raw per-device captures under uniform parsing, producing **AgriEdge**: 1,276,122 rows across five agricultural devices with full device attribution, 149,996 Modbus rows, and no column separating the classes above 0.0288. On this corrected benchmark, a leave-one-device-out sweep locates the generalisation boundary precisely: every detector but naive Bayes transfers between perception sensors with little loss (0.9106–0.9999 balanced accuracy), but collapses across the perception/actuation layer boundary, with random forest falling from 0.9988 to 0.5083 and logistic regression to 0.4877—chance level—when the Modbus gateway is withheld. Federated evaluation yields a further negative result: non-IID partitioning costs at most 0.0037 macro-F1, while the uplink required to train a 28,450-parameter model over LoRaWAN costs 4.6 hours. We release the audit tooling, the benchmark construction pipeline, and all experimental code.

*Keywords:* intrusion detection, industrial IoT, precision agriculture, data leakage, benchmark integrity, federated learning, edge computing, Edge-IIoTset

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## 1. Introduction

Precision agriculture has become one of the more consequential deployments of the industrial Internet of Things (IIoT). Soil-moisture probes, pH sensors, water-level floats and climate stations feed telemetry to edge gateways over lightweight publish/subscribe protocols, and those gateways in turn actuate irrigation pumps and valves over industrial fieldbus protocols such as Modbus TCP. The consequences of compromise are physical and immediate: a falsified soil-moisture reading can drive an irrigation controller to flood a field, and a denial-of-service attack against a message broker can prevent an irrigation trigger from firing during the narrow window in which it matters. Unlike an enterprise network, a farm has no on-site security staff, intermittent connectivity, and gateways that share a power budget with the machinery they control.

These constraints have made machine-learning intrusion detection an attractive proposition, and the Edge-IIoTset dataset of Ferrag et al. [1] has become the dominant vehicle for evaluating it. The dataset is unusually well-suited to the task on paper: it was captured from a physical seven-layer testbed with real sensors, it spans fourteen attack classes across MQTT, Modbus, HTTP and DNS, and it is explicitly partitioned to support both centralised and federated learning. It is also convenient, at 1.2 GB in its curated form, and it ships with a `Readme.txt` giving a step-by-step preprocessing recipe.

The results reported on it are, uniformly, extraordinary. Recent work reports 99.27% accuracy with 99.21% F1 [2] and 99.94% accuracy [3]. Such numbers are ordinarily read as evidence that the problem is close to solved and that research attention should move to deployment concerns—model compression, federated aggregation, inference latency.

This paper argues that the numbers should instead be read as a warning. We began with the intention of building a precision-agriculture intrusion detection system on Edge-IIoTset, following the standard pipeline. Before training, we audited the features the dataset's own documentation instructs researchers to construct. What we found is that the recipe manufactures a near-perfect label proxy out of a serialisation artifact.

### *1.1. The artifact in brief*

Edge-IIoTset encodes an absent protocol field as a zero rather than as a null—reasonable, since a packet carrying no MQTT layer has no MQTT topic. The normal-traffic captures and the attack captures were, however, parsed separately and concatenated afterwards, and the two branches serialised that zero differently: as the string `0` in one and as the string `0.0` in the other. Step 5 of the distributed recipe then applies `pandas.get_dummies` to seven columns containing these placeholders. Dummy encoding treats `0` and `0.0` as distinct tokens, so the resulting binary features encode *which file a row came from.* Because normal traffic and attack traffic came from different files, file provenance is the label.

The consequence is that four of the seven columns in the recipe recover the binary label with an accuracy of 1.0000 in isolation. No model that consumes those features is doing intrusion detection; it is reading a build artifact.

### *1.2. Contributions*

1. **We identify and characterise a label-leakage mechanism in the canonical Edge-IIoTset preprocessing recipe**, localise it to

placeholder serialisation, and quantify it with three independent measures (token purity, single-column held-out accuracy, normalised mutual information). Four columns reach a separation rate of 1.0000 (Section 5.1).

2. **We measure the artifact's contribution to reported performance** under 5-fold $\times$ 3-repeat cross-validation. Five of six standard classifiers reach exactly $1.0000 \pm 0.0000$ accuracy in every fold and the sixth reaches 0.99998; under the corrected protocol the same models span 0.6995 to 0.9503 macro-F1. A 1D-CNN and a deep MLP, evaluated separately on a single stratified split, reach 1.0000 accuracy under the same recipe and lose more than 0.21 macro-F1 under correction (Section 5.2).
3. **We show the leak is encoding-agnostic.** One-hot, label, ordinal and frequency encoding all yield identical 1.0000 single-column accuracy, because each is a bijection on the token set. The exposed population is therefore every study that treated these columns as categorical, not only those that copied the distributed recipe (Section 5.2.3).
4. **We show the curated subsets cannot support agricultural research**, with Modbus effectively absent and device identity stripped, and we rebuild the benchmark from the raw captures under uniform parsing (Section 5.3, Section 5.4).
5. **We locate the generalisation boundary in agricultural IIoT detection.** A leave-one-device-out sweep over all five devices shows that transfer between perception sensors is near-free—every model but naive Bayes scores 0.9880–0.9999 balanced accuracy when soil moisture, water level or pH is withheld, and 0.9106–0.9385 for temperature and humidity—while transfer across the perception/actuation layer collapses to 0.4877–0.5083, straddling the trivial baseline, for two of the six (Section 5.5).
6. **We show that the federated literature is optimising the wrong constraint.** Non-IID partitioning costs at most 0.0037 macro-F1, while training a 28,450-parameter model over LoRaWAN costs 4.6 hours of uplink (Section 5.6).
7. **We characterise edge deployment cost**, finding a 342$\times$ inference-latency spread and a 171$\times$ model-size spread across models whose random-split accuracies are indistinguishable (Section 5.7).
8. **We release the complete pipeline**—audit tooling, benchmark construction, and experiments—so that the audit can be applied to other

datasets (Section 8).[1]

## 2. Background

### 2.1. Precision agriculture as a layered IIoT system

The precision-agriculture deployments that Edge-IIoTset's testbed models decompose into three layers, and the security properties of each differ sharply.

**The perception layer** comprises the sensors that observe the physical field: soil moisture, pH, water level, temperature and humidity. Their traffic is low-rate, highly periodic, and semantically narrow—a soil-moisture probe emits a scalar at a fixed interval. This regularity is what makes anomaly detection tractable at this layer, and also what makes the layer a target: an adversary who can alter a reading controls the actuation logic downstream without ever touching the actuator.

**The network layer** carries that telemetry. MQTT dominates sensor reporting because its publish/subscribe model and small header suit constrained devices, while Modbus TCP remains the lingua franca for actuation because it is what irrigation controllers and PLCs speak. Neither protocol authenticates or encrypts by default in typical farm deployments. A man-in-the-middle positioned between a probe and the broker can rewrite a moisture reading; a denial-of-service attack against the broker prevents irrigation triggers from firing at all.

**The edge system layer** performs local processing. Agricultural sites are often served by cellular or LoRaWAN backhaul with severely constrained uplink, so detection must run on the gateway rather than in the cloud. This is the layer at which the tension between detection quality and resource budget binds.

### 2.2. The Edge-IIoTset dataset

Edge-IIoTset [1] was captured from a physical testbed spanning IoT sensors, edge gateways, SCADA components and adversary hosts. It is distributed in three parts: raw per-device normal-traffic captures (CSV and PCAP), raw per-attack captures, and two curated subsets—`ML-EdgeIIoT-dataset.csv` (157,800 rows) and `DNN-EdgeIIoT-dataset.csv` (2,219,201 rows)—intended for traditional machine learning and deep

[1] `https://github.com/MostafaGalal1/agriedge`

learning respectively. Both curated subsets carry 63 columns: 61 protocol features, a binary `Attack_label`, and a 15-class `Attack_type`.

Nearly all published work uses the curated subsets, and follows the `Readme.txt` recipe: drop fifteen identifier and payload columns, drop nulls and duplicates, and dummy-encode seven categorical columns. That recipe is the object of our audit.

### *2.3. Related work*

*Performance on Edge-IIoTset.* The literature is dense and its results are tightly clustered near ceiling. Reported figures include 99.27% accuracy with 99.21% F1 across four benchmark datasets [2] and 99.94% accuracy with a confusion matrix containing no errors of either kind [3]. Architectures range from autoencoder-based lightweight feature learning [3] to hybrid CNN-DNN models and TinyML deployments for energy-aware edge inference [4].

*Emerging scepticism.* A parallel literature has begun questioning whether these numbers mean what they appear to. Hakim et al. [5] train four lightweight architectures on one IIoT dataset and evaluate them, without retraining, on two structurally distinct ones; they find that models rely on coarse port-category shortcuts appearing 96–435× more frequently in source-domain attacks, and that prior work evaluates models only within their training network, leaving behaviour on unseen networks unverified. A dataset-centric review in *Frontiers in Big Data* [6] catalogues evaluation biases across IoT/IIoT intrusion detection, noting that published results can contain undocumented preprocessing operations, hidden leakage mechanisms, or inconsistent validation protocols, and that no studies report $k$-fold cross-validation on Edge-IIoTset, relying instead on 80/20 splits. Recent work on cross-domain heterogeneous benchmarking [7] and dataset-centric evaluation of federated intrusion detection [8] points in the same direction.

*Benchmark integrity as a research programme.* A distinct line of work audits intrusion detection benchmarks directly rather than proposing models for them. Engelen et al. [9] reverse-engineer CICIDS2017 and find labelling and traffic-generation faults severe enough to change reported conclusions. Flood et al. [10] generalise this into a catalogue of six recurring *data design smells* obtained by manual analysis of seven highly-cited benchmark datasets, among them *wrong label*, *traffic collapse* and—closest to our finding—*highly dependent features*, where distinct features carry near-identical information about the label. Arp et al. [11] survey thirty top-tier security papers and codify the recurring methodological pitfalls of learning-based security, spurious correlation among them.

This paper sits inside that programme rather than alongside it, and two things distinguish its contribution. First, Edge-IIoTset is not among the seven datasets Flood et al. examine, and we are not aware of any comparable audit of it; the catalogue therefore does not yet cover the dataset that IIoT and agricultural work overwhelmingly relies on. Second, the mechanism differs in kind. The dependent features documented in that literature are genuine network properties—destination port, packet size, flow duration—that happen to correlate with the label because of how traffic was generated. The columns we identify are not network properties at all under the values that carry the signal: the discriminating token is the placeholder written where a protocol field was *absent*, and its spelling records which file the row was parsed from. The dependency is thus perfect rather than merely strong, and it is invisible to any inspection that casts the column to a numeric type before looking at it.

Two results are closer still to ours. Bouke and Abdullah [12] study *pattern leakage* introduced during preprocessing rather than present in the raw capture, and show empirically that it inflates reported reliability—establishing the general phenomenon that this paper instantiates in a specific dataset with a specific mechanism. Kostas et al. [13] show that individual-packet features let models memorise artifacts that do not survive transfer to another network, which is the failure mode our leave-one-device-out sweep exhibits at a layer boundary (Section 5.5). Neither reports a label-recovering artifact in Edge-IIoTset, and to our knowledge no published work does.

*Our position relative to this work.* Two literatures meet here. The IIoT-specific sceptical work has established *that* generalisation fails and *that* leakage is suspected, without localising a mechanism. The benchmark-integrity work has localised mechanisms precisely, but in the classical network-intrusion datasets rather than in Edge-IIoTset. This paper joins the two: it applies the auditing stance of the latter to the dataset the former is built on, and finds a mechanism that is both stronger than the dependencies previously catalogued—it recovers the label alone, at accuracy 1.0000—and different in kind, since the discriminating value encodes file provenance rather than any property of the traffic. Our finding is complementary to Hakim et al.'s: they show models learn shortcuts that fail to transfer; we show that for Edge-IIoTset under its own documented recipe, the shortcut is not a network feature at all.

*Federated learning for agricultural IIoT.* Federated approaches have been proposed specifically for agricultural intrusion detection, including secure federated deep reinforcement learning [14]. These evaluations typically partition data uniformly at random, producing independent and identically distributed (IID) clients. Section 5.6 shows why that choice matters.

## 3. Threat Model

We consider an adversary with network access to a farm's IIoT segment, obtained through a compromised gateway, an exposed cellular modem, or physical proximity to an unsecured wireless link. The adversary cannot compromise the sensors' physical measurements but can observe, inject, and modify network traffic. We assume the detector runs on the edge gateway and that the gateway itself is trusted; detection of a compromised gateway is out of scope.

The attack classes in Edge-IIoTset map onto agricultural consequences as follows. **Man-in-the-middle** (ARP and DNS spoofing) permits alteration of telemetry in flight—the falsified-moisture scenario. **Denial of service** in its four variants (TCP SYN, UDP, ICMP, HTTP floods) severs the telemetry path, preventing irrigation triggers. **Reconnaissance** (port scanning, OS fingerprinting, vulnerability scanning) precedes targeted attacks. **Injection and application attacks** (SQL injection, XSS, file upload) target the management interfaces through which farm operators configure irrigation schedules. **Backdoor and ransomware** establish persistence, with ransomware against an irrigation controller carrying an unusually compressed decision timeline given crop water stress.

Two of these classes are severely under-represented in the dataset—MITM at 1,229 rows and OS fingerprinting at 1,001—despite MITM being the most consequential for the falsified-telemetry scenario. We preserve this natural rarity rather than resampling it away, and report per-class recall accordingly.

## 4. Methodology

### *4.1. Auditing for label leakage*

Our audit asks one question per categorical column: *how much of a classifier's apparent skill can this column supply on its own, without any network behaviour being modelled?* We compute three measures so that no single statistic carries the argument.

*Token purity.* For each distinct token in a column, we count occurrences under each label. A token is *pure* if it occurs under exactly one label. The **separation rate** is the fraction of rows covered by pure tokens. A column with a separation rate of 1.0 is a relabelling of the target.

*Single-column held-out accuracy.* We fit a decision tree to the one-hot encoding of that column alone, using a stratified 80/20 split, and report test accuracy. This measures directly what a model can extract from the column.

*Normalised mutual information.* between the column's tokens and the label, which is estimator-free and scale-free.

We additionally run a mechanism-specific probe. For each column containing both the token `0` and the token `0.0`, we tabulate the label distribution of each spelling separately. If each spelling occurs under exactly one label, the column carries a **provenance marker**: the spelling of a value denoting *absence*—which has no network semantics whatsoever—determines the label with certainty.

Columns audited are exactly the seven that `Readme.txt` Step 5 instructs researchers to dummy-encode:

> `http.request.method`, `http.referer`, `http.request.version`, `dns.qry.name.len`, `mqtt.conack.flags`, `mqtt.protoname`, `mqtt.topic`.

Critically, we read all columns as strings. Pandas' type inference silently normalises `0` and `0.0` to the same float, which destroys the artifact before it can be observed—a plausible reason it has gone unreported.

### *4.2. The two preprocessing protocols*

*Protocol A (as distributed).* reproduces `Readme.txt` Steps 4 and 5 verbatim: drop the fifteen listed columns, drop rows with nulls, drop duplicates, dummy-encode the seven listed columns, and coerce the remainder to numeric.

*Protocol B (corrected).* applies four changes:

1. **Placeholder canonicalisation.** Every spelling denoting absence (`0`, `0.0`, empty, `nan`, `None`, `null`) is collapsed to a single sentinel before encoding. This removes the provenance signal while preserving the field's semantic content: *this protocol layer was not present in this packet.*

2. **Deduplication before splitting.** Exact duplicates spanning a train/test boundary are memorised rather than generalised.
3. **Structural rather than identity encoding of high-cardinality strings.** Columns such as `http.request.version` contain whole injected request lines on the attack side. One-hot encoding these lets a model memorise literal payload strings, which neither transfers to another network nor survives a single changed byte. We replace token identity with structural properties—length, character-class ratios, Shannon entropy, and presence of injection-associated syntax.
4. **Post-hoc removal of residual separators.** Any column still separating the classes above a configurable threshold (default 0.999) after the above is dropped.

### *4.3. Constructing the AgriEdge benchmark*

The curated subsets cannot support an agricultural study (Section 5.3). We therefore rebuild from the raw per-device captures, which retain full device attribution. All raw captures share a byte-identical 63-column header, which makes uniform construction possible.

Two properties are enforced. **Uniform parsing**: every capture—normal and attack alike—is read with identical dtype handling and identical placeholder canonicalisation applied across *all* columns. Because the artifact arises from concatenating differently-parsed frames, parsing uniformly makes it structurally impossible rather than merely correcting it after the fact. **Preserved device attribution**: each row records the device or attack capture it came from, which is what makes non-IID federated partitioning by farm possible, and which the curated subsets discard.

Sampling is per-chunk at a constant rate rather than by prefix, which is unbiased with respect to position in the capture session. Normal devices are capped at 150,000 rows each and attack classes at 60,000; because sampling is per chunk, a class may overshoot its cap by less than one chunk. Classes smaller than the cap are taken whole, preserving the natural rarity of MITM and fingerprinting. Host identifiers and raw payload columns are dropped at build time so no downstream protocol can reintroduce them.

### *4.4. Evaluation protocols*

We evaluate under two splitting regimes. **Random stratified** (80/20) is the convention in the literature and provides comparability. **Leave-one-device-out (LODO)** withholds all normal traffic from one sensor type and tests on it, with 20% of attack traffic. This models the deployment reality that a farm adds a sensor the detector was never trained on. It is a genuine distribution shift, and the closest within-dataset analogue of the cross-network evaluation that Hakim et al. [5] argue is missing from the literature.

### *4.5. Model suite*

We use six estimators chosen to mirror those most frequently reported on Edge-IIoTset: decision tree, random forest, histogram gradient boosting, logistic regression, Gaussian naive Bayes, and a two-hidden-layer MLP (64, 32). Every factory returns a fresh estimator; none is shared between protocols, folds, or federated clients. All randomness is seeded.

The deliberate inclusion of weak models is methodological. If a classifier that cannot represent the decision boundary nonetheless achieves perfect accuracy, the features—not the classifier—are doing the work. Gaussian naive Bayes serves as this canary throughout.

### *4.6. Federated and edge-cost evaluation*

Federated experiments use FedAvg [15] over a compact MLP sized for a gateway, comparing three client constructions: IID (the prior-work baseline), one client per sensor type, and one client per farm, where farms group devices as {soil moisture, water level}, {temperature-humidity, pH}, and {Modbus}. Clients exchange only parameters.

Edge cost is measured as serialised model size, parameter count, and inference latency reported as median and 95th percentile over 30 timed single-sample batches. We report the tail because a late verdict is a missed actuation window. Communication cost is computed against representative rural backhaul profiles: LoRaWAN SF7 (5.47 kbit/s), NB-IoT (250), LTE Cat-M1 (1,000), and rural 4G (5,000).

## 5. Results

### *5.1. Four columns recover the label perfectly*

Table 1 reports the audit of the seven columns named in the distributed recipe, on the ML subset (157,800 rows; 24,301 normal, 133,499 attack).

Four columns separate every row in the dataset. A decision tree given one of these columns and nothing else achieves perfect held-out accuracy. Normalised mutual information with the label reaches 0.9859.

Table 2 isolates the mechanism.

Table 1: Leakage audit of the columns `Readme.txt` Step 5 instructs researchers to dummy-encode (ML subset).

| Column | Tokens | Pure normal | Pure attack | Rows separated | Separation rate | 1-col. accuracy | NMI |
|---|---|---|---|---|---|---|---|
| `dns.qry.name.len` | 8 | 6 | 2 | 157,800 | **1.0000** | **1.0000** | 0.9859 |
| `mqtt.conack.flags` | 3 | 2 | 1 | 157,800 | **1.0000** | **1.0000** | 0.9642 |
| `mqtt.protoname` | 3 | 2 | 1 | 157,800 | **1.0000** | **1.0000** | 0.9649 |
| `mqtt.topic` | 3 | 2 | 1 | 157,800 | **1.0000** | **1.0000** | 0.9650 |
| `http.request.version` | 8 | 0 | 7 | 62,472 | 0.3959 | 0.8460 | 0.1378 |
| `http.request.method` | 6 | 0 | 5 | 61,258 | 0.3882 | 0.8460 | 0.1348 |
| `http.referer` | 4 | 0 | 3 | 30,689 | 0.1945 | 0.8460 | 0.0783 |

Table 2: Label distribution of the two spellings of the zero placeholder (ML subset).

| | `'0'` | | `'0.0'` | |
|---|---|---|---|---|
| Column | Normal | Attack | Normal | Attack |
| `dns.qry.name.len` | 24,272 | **0** | **0** | 133,272 |
| `mqtt.conack.flags` | 23,012 | **0** | **0** | 133,499 |
| `mqtt.protoname` | 23,051 | **0** | **0** | 133,499 |
| `mqtt.topic` | 23,055 | **0** | **0** | 133,499 |
| `http.request.method` | **0** | 54,062 | 24,301 | 72,241 |
| `http.referer` | **0** | 30,399 | 24,301 | 102,810 |
| `http.request.version` | **0** | 55,276 | 24,301 | 71,027 |

The pattern is unambiguous. In the four fully-separating columns, `'0'` never appears in an attack row and `'0.0'` never appears in a normal row. The three HTTP columns show the same artifact with the polarity reversed—`'0'` appears only in attack rows—separating 19–40% of the data.

This is a build artifact, not a network phenomenon. Both tokens denote the same thing: the protocol layer was absent. Their distinction records only which parsing branch wrote the row. We additionally measured duplicate rows, which inflate accuracy under random splitting: 814 exact duplicates (0.52%), all with consistent labels.

*The artifact is not confined to the small subset.* Because deep-learning studies use the larger `DNN-EdgeIIoT-dataset.csv`, we repeated the audit there. Table 3 shows the identical mechanism at fourteen times the scale—and on a subset whose class balance is *inverted* relative to the ML subset (1,615,643 normal against 603,558 attack), ruling out any explanation contingent on which class is the majority.

Table 3: The same four columns on the 2,219,201-row DNN subset.

| | Separation rate | 1-col. accuracy | NMI | `'0'` normal / attack | `'0.0'` normal / attack |
|---|---|---|---|---|---|
| `dns.qry.name.len` | **1.0000** | **1.0000** | 0.9927 | 1,613,798 / **0** | **0** / 603,331 |
| `mqtt.conack.flags` | **1.0000** | **1.0000** | 0.8879 | 1,532,586 / **0** | **0** / 603,558 |
| `mqtt.protoname` | **1.0000** | **1.0000** | 0.8881 | 1,532,608 / **0** | **0** / 603,558 |
| `mqtt.topic` | **1.0000** | **1.0000** | 0.8881 | 1,532,627 / **0** | **0** / 603,558 |

Every one of 2,219,201 rows is separated, by each of four columns independently. Both branches of the Edge-IIoTset literature—the classical-ML branch using the ML subset and the deep-learning branch using the DNN subset—are therefore exposed to the same artifact.

### *5.2. What the artifact is worth*

Because a single split cannot distinguish an effect from split variance—and because the review we cite [6] notes that no published study reports $k$-fold on this dataset—we evaluate under 5-fold $\times$ 3-repeat stratified cross-validation, reporting mean and 95% interval over all fifteen folds (Table 4).

Three observations.

**Five of the six models attain exactly** $1.0000 \pm 0.0000$**, in every one of fifteen folds.** Not 0.999, and not on a lucky split: zero variance across fifteen independent stratified partitions, for five structurally different learners. The sixth, the small (64, 32) MLP, averages 0.99998—fewer than

Table 4: Binary classification under the distributed recipe (A) versus the corrected protocol (B). ML subset, 15 folds, mean ± 95% CI half-width. Five of the six models score exactly 1.0000±0.0000 under the recipe; the MLP row is reported to five decimals because it does not, and rounding it to four would conceal that.

| Model | Accuracy (A) | Accuracy (B) | Macro-F1 (A) | Macro-F1 (B) | ΔF1 |
|---|---|---|---|---|---|
| Hist. gradient boosting | **1.0000 ± 0.0000** | 0.9752 ± 0.0005 | **1.0000 ± 0.0000** | **0.9503 ± 0.0011** | 0.0497 |
| Random forest | **1.0000 ± 0.0000** | 0.9658 ± 0.0005 | **1.0000 ± 0.0000** | 0.9343 ± 0.0009 | 0.0657 |
| Decision tree | **1.0000 ± 0.0000** | 0.9610 ± 0.0005 | **1.0000 ± 0.0000** | 0.9271 ± 0.0009 | 0.0729 |
| MLP | 0.99998 ± 0.00001 | 0.9095 ± 0.0008 | 0.99996 ± 0.00003 | 0.7858 ± 0.0023 | 0.2142 |
| Logistic regression | **1.0000 ± 0.0000** | 0.8993 ± 0.0008 | **1.0000 ± 0.0000** | 0.7567 ± 0.0024 | 0.2433 |
| Gaussian naive Bayes | **1.0000 ± 0.0000** | 0.8742 ± 0.0125 | **1.0000 ± 0.0000** | 0.6995 ± 0.0089 | **0.3005** |

one misclassified row per fold out of 31,560, and the only model in the suite whose recipe score is not bit-exact. We report it unrounded rather than let it round up to the same 1.0000 as the rest, since a paper about numbers that were never checked closely should survive being checked closely itself. Perfect or near-perfect invariant separation of a real network-traffic capture is not a plausible outcome of intrusion detection. It is the signature of a feature that *is* the label.

**The canary fires.** Gaussian naive Bayes assumes conditional independence of features given the class and cannot represent the interactions a genuine intrusion-detection boundary requires. Its perfect, zero-variance score under the distributed recipe is the clearest evidence available that the features contain a direct label encoding. Under correction it falls 0.3005 macro-F1, and its interval widens by an order of magnitude relative to the other models (±0.0089 against ±0.0009 for the trees)—the behaviour of a model that is genuinely struggling, as it should be.

**Honest ceiling.** Under correction the best model reaches 0.9503±0.0011 macro-F1—respectable, and roughly four points below the above-99% range the literature reports.

*5.2.1. Deep models are equally deceived*

A reviewer may reasonably object that the literature's headline numbers come from deep architectures, and that a deep model might have been robust to the artifact. It is not. Table 5 trains a deep MLP (61,378 parameters under the distributed recipe, 59,586 under the corrected one, the difference being input width) and a 1D-CNN (6,658 parameters)—the latter matching the construction used in several published Edge-IIoTset papers—under both protocols on a single identical stratified split. This is a separate experiment from the fifteen-fold comparison in Table 4, whose MLP is the smaller (64, 32) estimator from the classical suite.

Table 5: Deep baselines under both protocols (ML subset, 15 epochs).

| Architecture | Accuracy (A) | Accuracy (B) | Macro-F1 (A) | Macro-F1 (B) | ΔF1 |
|---|---|---|---|---|---|
| 1D-CNN | **1.0000** | 0.9003 | **1.0000** | 0.7598 | 0.2402 |
| MLP | **1.0000** | 0.9093 | 0.9999 | 0.7848 | 0.2151 |

Both reach 1.0000 accuracy under the distributed recipe, and both lose more than 0.21 macro-F1 under correction—a larger drop than any tree-based model. Depth confers no protection whatsoever. A network consuming a leaked one-hot feature reads it exactly as a decision stump does, and its additional capacity is spent fitting a signal that will not exist at deployment.

*5.2.2. The effect holds at scale on the deep-learning subset*

Table 6 repeats the protocol comparison on the DNN subset. After cleaning, 1,909,671 rows remain; the corrected protocol rewrote 15,218,483 placeholder cells and removed 309,530 duplicates.

Table 6: Protocol comparison on the DNN subset (1,909,671 rows).

| Model | Accuracy (A) | Accuracy (B) | Macro-F1 (A) | Macro-F1 (B) | ΔF1 |
|---|---|---|---|---|---|
| Hist. gradient boosting | **1.0000** | **0.9694** | **1.0000** | **0.9612** | 0.0388 |
| Random forest | **1.0000** | 0.9596 | **1.0000** | 0.9499 | 0.0501 |
| Decision tree | **1.0000** | 0.9502 | **1.0000** | 0.9390 | 0.0610 |
| MLP | **1.0000** | 0.9141 | **1.0000** | 0.8832 | 0.1168 |
| Logistic regression | **1.0000** | 0.8948 | **1.0000** | 0.8562 | 0.1438 |
| Gaussian naive Bayes | **1.0000** | 0.5203 | **1.0000** | 0.5190 | **0.4810** |

Every model reaches 1.0000 accuracy on 1.9 million rows under the distributed recipe. The ordering under correction is identical to the ML subset, and naive Bayes again collapses—here to 0.5190 macro-F1, essentially the performance of a coin flip on a task it appeared to have solved perfectly. That the effect reproduces across two subsets differing by an order of magnitude in size, with inverted class balance, rules out sampling accident.

One difference is worth noting: the corrected DNN subset supports slightly *higher* scores than the corrected ML subset (0.9612 against 0.9503 macro-F1 for the best model), the expected consequence of fourteen times more training data. Correcting the artifact does not merely lower scores; it restores the ordinary relationship between dataset size and achievable performance that a saturated benchmark conceals.

### *5.2.3. The leak is encoding-agnostic*

The distributed recipe uses `pd.get_dummies`, and it would be convenient to conclude that only studies adopting that specific call are affected. They are not. The signal is the *distinction between the tokens* `'0'` and `'0.0'`; any encoding that preserves that distinction transmits it. Table 7 fits a decision tree to a single column under four encodings in common use.

Table 7: Single-column held-out accuracy by encoding scheme (ML subset).

| Column | One-hot (recipe) | Label / factorize | Ordinal | Frequency |
|---|---|---|---|---|
| `dns.qry.name.len` | **1.0000** | **1.0000** | **1.0000** | **1.0000** |
| `mqtt.conack.flags` | **1.0000** | **1.0000** | **1.0000** | **1.0000** |
| `mqtt.protoname` | **1.0000** | **1.0000** | **1.0000** | **1.0000** |
| `mqtt.topic` | **1.0000** | **1.0000** | **1.0000** | **1.0000** |
| `http.request.version` | 0.8460 | 0.8460 | 0.8460 | 0.8460 |
| `http.request.method` | 0.8460 | 0.8460 | 0.8460 | 0.8460 |
| `http.referer` | 0.8460 | 0.8460 | 0.8460 | 0.8460 |

Across the four leaking columns and all four encodings, the minimum single-column accuracy is 1.0000—the values are not merely similar but identical, because each encoding is a bijection on the token set and a decision tree is invariant to bijective relabelling of a categorical input.

This matters for scoping the problem. The affected population is not "studies that followed `Readme.txt` verbatim" but **studies that treated these columns as categorical at all**—which, given that four of them are MQTT and DNS protocol fields that any IIoT feature set would naturally include, is a far larger group. A study reporting label encoding rather than dummy encoding has not thereby avoided the artifact, and cannot be assumed unaffected.

### *5.3. The curated subsets cannot support agricultural research*

Two structural facts emerged during scoping.

**Modbus is absent.** Across all 157,800 rows of the ML subset, `mbtcp.len`, `mbtcp.trans_id` and `mbtcp.unit_id` are uniformly zero. In the DNN subset, 150 of 2,219,201 rows (0.0068%) carry a non-zero Modbus transaction ID. Any study claiming to model attacks on irrigation actuation using these subsets is not observing Modbus traffic.

**Device identity is stripped.** The only surviving value of `mqtt.topic` in either subset is `Temperature_and_Humidity` (1,246 rows in ML; 83,016 in DNN). Soil moisture, pH and water level—the sensors that define precision agriculture—carry no topic attribution. Filtering to “agricultural sensors” is therefore impossible on the curated data.

Both are recoverable from the raw captures, which retain 1,192,777 soil-moisture rows, 2,295,288 water-level rows, 1,615,722 temperature-humidity rows, 746,908 pH rows, and 159,502 Modbus rows.

### *5.4. AgriEdge: construction and validation*

Table 8 summarises the rebuilt benchmark: 1,276,122 rows, 49 features, 751,559 normal and 524,563 attack (attack rate 0.4111).

Table 8: AgriEdge composition.

| Layer | Source | Rows |
|---|---|---|
| Perception | pH value | 150,987 |
| Perception | Soil moisture | 150,341 |
| Perception | Water level | 150,187 |
| Perception | Temperature & humidity | 150,048 |
| Actuation | Modbus | 149,996 |
| Adversary | 14 attack classes | 524,563 |

Attack classes range from 60,069 (DDoS-HTTP) down to 1,229 (MITM) and 1,001 (fingerprinting), preserving natural rarity.

Table 9 re-runs the audit on the constructed benchmark. The comparison with Table 1 is the validation of the method.

Table 9: Leakage audit of AgriEdge, same columns as Table 1.

| Column | Separation rate | 1-col. accuracy | NMI | Provenance marker |
|---|---|---|---|---|
| `http.request.version` | 0.0288 | 0.6179 | 0.0641 | no |
| `http.request.method` | 0.0288 | 0.6179 | 0.0643 | no |
| `mqtt.conack.flags` | 0.0272 | 0.5889 | 0.0366 | no |
| `mqtt.topic` | 0.0272 | 0.5889 | 0.0349 | no |
| `mqtt.protoname` | 0.0270 | 0.5889 | 0.0363 | no |
| `dns.qry.name.len` | 0.0026 | 0.5891 | 0.0041 | no |
| `http.referer` | 0.0005 | 0.5894 | 0.0012 | no |

**No column separates the classes.** Maximum separation rate falls from 1.0000 to 0.0288; maximum single-column accuracy from 1.0000 to 0.6179; maximum NMI from 0.9859 to 0.0643. No provenance markers remain. Uniform parsing eliminates the artifact by construction.

### *5.5. Detection fails across layers, not across devices*

Table 10 evaluates on AgriEdge (400,000-row subsample, 409 encoded features), withholding each of the five devices in turn. For comparison, all six models score between 0.6914 and 0.9988 balanced accuracy on the same data under a conventional random stratified split.

Table 10: Balanced accuracy under leave-one-device-out, every device withheld in turn.

| Model | Soil moisture | Water level | pH value | Temp. & humidity | **Modbus** | Mean | Worst |
|---|---|---|---|---|---|---|---|
| MLP | 0.9997 | 0.9998 | 0.9994 | 0.9121 | **0.8164** | 0.9455 | 0.8164 |
| Hist. gradient boosting | 0.9997 | 0.9999 | 0.9995 | 0.9385 | **0.7464** | 0.9368 | 0.7464 |
| Decision tree | 0.9997 | 0.9999 | 0.9995 | 0.9161 | **0.6793** | 0.9189 | 0.6793 |
| Random forest | 0.9997 | 0.9998 | 0.9995 | 0.9118 | **0.5083** | 0.8838 | 0.5083 |
| Logistic regression | 0.9880 | 0.9885 | 0.9880 | 0.9106 | **0.4877** | 0.8726 | 0.4877 |
| Gaussian naive Bayes | 0.6153 | 0.6248 | 0.6158 | 0.6022 | **0.8068** | 0.6530 | 0.6022 |

The full sweep **refines, and partly overturns, the conclusion a single held-out device would have supported**. Had we withheld only Modbus, we would have reported that introducing any unseen sensor collapses detection. That is not what happens.

**Perception sensors are mutually substitutable.** Withholding soil moisture, water level or pH costs almost nothing: every model except naive Bayes stays in the 0.9880–0.9999 band, and the tree-based models remain at 0.9995 or better. A detector trained on three agricultural sensors transfers to a fourth essentially without loss. Temperature and humidity is slightly harder (0.9106–0.9385), consistent with it being the one device that carries a distinct MQTT topic in the original capture.

**The failure is across layers, not across devices.** Only Modbus—the sole actuation-layer device, speaking a different protocol—produces collapse: random forest 0.5083, logistic regression 0.4877, straddling the trivial baseline of 0.5. The generalisation boundary in this benchmark is the perception/actuation layer boundary, not device identity.

This is a sharper and more useful claim than the one a single holdout would have licensed. It says where a practitioner may safely extrapolate—across sensors of similar protocol profile—and where they may not—across an architectural layer. It also refines Hakim et al.'s [5] cross-network result by locating the shift that matters: not merely "a different network," but a different protocol stratum within the same network.

**The ranking inversion is real but layer-specific.** Gaussian naive Bayes is by a wide margin the worst model on every perception holdout (0.6022–0.6248) and the *second best* on Modbus (0.8068)—the only model that improves under layer shift. Random forest is the mirror image: joint-best on perception holdouts (0.9997) and worst on Modbus (0.5083). Selecting by mean or by random-split performance would pick random forest; selecting for worst-case robustness picks the MLP, which is simultaneously best on mean (0.9455) and best on the hard case (0.8164). A practitioner optimising average performance would not select the model that survives the shift that actually occurs.

**Random splitting remains optimistic in the aggregate.** Four models reach 0.9988 balanced accuracy under random splitting, above their worst-case LODO performance by 0.18 to 0.49. This is not leakage in the Section 5.1 sense—the audit confirms no column separates the classes—but random splitting places packets from the same capture session and attack burst on both sides of the boundary. Random-split accuracy on AgriEdge should be read as an upper bound.

### *5.6. Federated learning: heterogeneity is cheap, the uplink is not*

Table 11 reports FedAvg over a 28,450-parameter MLP after 20 communication rounds on a 300,000-row sample, under three client constructions.

Table 11: Global model after 20 FedAvg rounds by client construction.

| Client construction | Clients | Accuracy | Balanced accuracy | Macro-F1 |
|---|---|---|---|---|
| IID (prior-work baseline) | 5 | 0.9970 | 0.9968 | 0.9969 |
| Per-farm | 3 | 0.9971 | 0.9970 | 0.9970 |
| Per-device | 5 | 0.9934 | 0.9939 | 0.9932 |

The result is a negative one, and we report it as such. **Heterogeneity costs remarkably little.** Partitioning one client per sensor type—the maximally heterogeneous construction, in which each client observes a single protocol profile—costs only 0.0037 macro-F1 relative to IID. Per-farm partitioning is indistinguishable from IID.

Heterogeneity does slow convergence: the per-device split reaches 0.9932 macro-F1 only at round 19, while the per-farm split passes 0.9950 by round 8 and plateaus at round 12. Under a fixed round budget the gap would be larger. But the common expectation that non-IID partitioning substantially degrades final federated accuracy is not supported here. We caution that this conclusion inherits the limitation of Section 5.5: it is measured under random splitting of the test set.

**The binding constraint is the uplink, not the statistics.** Table 12 reports the communication cost of the same model against rural backhaul profiles, assuming five clients uploading dense parameter vectors.

On LoRaWAN—a common choice precisely because agricultural sites are remote—a single round costs nearly fourteen minutes of pure uplink, and a

Table 12: Uplink cost of federated training (28,450 parameters, 5 clients, 20 rounds).

| Backhaul | Uplink (kbit/s) | Seconds per round | Minutes for full training |
|---|---|---|---|
| LoRaWAN SF7 | 5.47 | **832.2** | **277.4** |
| NB-IoT | 250 | 18.2 | 6.1 |
| LTE Cat-M1 | 1,000 | 4.6 | 1.5 |
| Rural 4G | 5,000 | 0.9 | 0.3 |

20-round training run costs **4.6 hours** of continuous transmission, before accounting for duty-cycle regulations that cap LoRaWAN transmit time far below 100%. Federated learning in this configuration is not deployable over LoRaWAN at all.

This reframes the design problem. The literature's emphasis on handling non-IID data is, on this benchmark, addressing the smaller of the two obstacles. A farm gateway can tolerate the 0.0037 macro-F1 that heterogeneity costs; it cannot tolerate 4.6 hours of uplink. Work on gradient compression, sparsification, or reduced round counts would buy far more deployability here than work on heterogeneity-robust aggregation.

### *5.7. Edge deployment cost*

Table 13 reports deployment footprint for the five models plausible on a gateway.

Table 13: On-gateway deployment cost (single-sample inference, 30 timed repeats).

| Model | Size (KiB) | Parameters | Median latency ($\mu$s) | p95 latency ($\mu$s) | Throughput (/s) |
|---|---|---|---|---|---|
| Decision tree | **20.9** | 109 | **40.1** | **52.1** | **24,935** |
| Logistic regression | 25.1 | 410 | 108.9 | 123.3 | 9,183 |
| Gaussian naive Bayes | 34.5 | 1,636 | 108.9 | 131.0 | 9,185 |
| Hist. gradient boosting | 388.1 | — | 4,731.9 | 6,307.6 | 211 |
| Random forest | **3,563.6** | 45,002 | **13,738.8** | 14,215.1 | **73** |

The spread is severe among models whose random-split accuracies are statistically indistinguishable. Random forest and decision tree both score 0.9988 balanced accuracy under random splitting, yet random forest is **342× slower** and **171× larger**. Its 13.7 ms median inference gives a throughput of 73 samples/second—below the aggregate packet rate of even a modest farm deployment, meaning it cannot run inline.

Read together with Table 10, the practical conclusion is sharp. Random forest is simultaneously the most expensive model to deploy *and* the one that fails hardest across the layer boundary. Its apparent 0.9986 macro-F1 under random splitting is the least informative number in this study. The decision tree, at 1/171 the size and 1/342 the latency, is strictly preferable on both axes.

## 6. Discussion

### *6.1. What this means for the Edge-IIoTset literature*

We do not claim that every published result on Edge-IIoTset is invalid. Studies that operate on raw captures, that use only numeric protocol features, or that exclude the four affected columns are unaffected. But the affected recipe is the one the dataset authors distribute, and the columns are among the most intuitively appealing features in the dataset—`mqtt.topic` and `mqtt.protoname` look like exactly the MQTT-layer features an IIoT intrusion detector ought to use.

*How far has the recipe propagated?.* We surveyed public implementations and papers for the recipe's fingerprints. Three observations follow.

First, the recipe is reproduced verbatim in public code. The helper function `encode_text_dummy`—an unusual name, traceable to Jeff Heaton's teaching material and carried into the dataset's `Readme.txt`—appears in public Edge-IIoTset implementations together with the exact fifteen-column drop list and the exact seven dummy-encoded columns, including in a CNN-LSTM implementation. Researchers are copying the recipe, not merely reading it.

Second, and more consequentially, **avoiding `get_dummies` does not avoid the artifact.** Section 5.2.3 shows the leak survives label, ordinal and frequency encoding unchanged. A study that reports "categorical variables were converted using label encoding" has not escaped it.

Third, the diagnostic signature is visible in published results. Hasan et al. [3], who report label encoding rather than dummy encoding, publish a confusion matrix in which their best model produces 4,820 true positives, 25,620 true negatives, zero false positives and zero false negatives—a perfect classification of the held-out set—alongside 99.94% accuracy. We do not have access to their exact column list and so cannot attribute this with certainty. We observe only that a flawless confusion matrix on real network traffic is precisely what the artifact produces, that their stated encoding preserves it, and that their reported correlation-based feature pruning (dropping features correlated above 0.6) would collapse the four mutually-redundant leaking columns to one—which Table 1 shows is sufficient on its own.

We raise this not to single out one paper. The recipe was distributed by the dataset's authors, the affected columns are the natural ones to use, and the artifact is invisible to anyone who loads the CSV with default pandas type inference. Following the documentation was the reasonable thing to do. The point is that the resulting numbers cannot be interpreted as detection performance, and that the affected population is much larger than a citation search for the recipe would suggest.

### *6.2. Why the artifact survived*

Three factors plausibly explain why an artifact this large went unreported for four years.

First, **pandas hides it**. Default type inference coerces `0` and `0.0` to an identical float. A researcher who loads the CSV normally, inspects the columns, and finds them numerically identical has no reason to suspect anything. The artifact is visible only when columns are read as strings—which is exactly what `get_dummies` effectively does downstream, after the encoding decision has already been made.

Second, **the result looks like success**. A pipeline returning 99.9% accuracy does not prompt debugging. The incentive to investigate an anomalously good result is weaker than the incentive to investigate a bad one.

Third, **the recipe carries the dataset's authority**. It ships in the dataset's own `Readme.txt`. Following it is the conservative choice, and deviation would require justification.

### *6.3. Implications for precision-agriculture deployment*

*Select on worst-case, not average, cross-layer performance.* Table 10 shows random forest is joint-best on perception holdouts and worst on the actuation holdout. Because the perception cases are nearly free for every model, average LODO performance is dominated by the easy cases and hides exactly the failure that matters. Evaluate against a held-out device from a *different architectural layer* before selecting.

*A detector validated on sensors does not transfer to actuators.* This is the operationally important consequence. A farm that validates its IDS across its soil, water and pH probes will see excellent numbers and may reasonably conclude the detector generalises. Adding a Modbus irrigation controller then places it in the regime where a third of the model suite performs at chance. Actuation-layer traffic needs its own validation, and arguably its own detector.

*Prefer simpler and smaller than the literature implies.* The decision tree is 342× faster and 171× smaller than the random forest while degrading far less across the layer boundary (0.6066 against 0.3098 macro-F1). The MLP is best on both mean (0.9455) and worst case (0.8164) at moderate cost, and is our recommended default.

*Treat MITM detection as an open problem.* The class that matters most for falsified telemetry is represented by 1,229 rows. No result in this paper should be read as evidence that MITM detection in agricultural IIoT is solved.

### *6.4. Generality of the failure mode*

The mechanism—a placeholder serialised differently in two branches of a dataset build, then encoded—is not specific to Edge-IIoTset. It requires only that a dataset be assembled by concatenating separately-parsed sources whose separation correlates with the label, which describes a large fraction of security datasets, where benign and malicious traffic are captured in separate sessions. We would expect the audit to find similar artifacts elsewhere, and we release it as a general tool for that reason.

## 7. Threats to Validity

*Construct validity.* Our separation-rate measure treats a column as leaking if its tokens are label-pure. In a genuinely separable problem, a legitimately predictive feature could be pure without being an artifact. We address this with the provenance probe, which tests the specific hypothesis that the *spelling of an absence marker* carries the signal—a quantity with no network semantics under any reading. The four flagged columns all satisfy this stricter test.

*Internal validity.* Protocols A and B differ in more than placeholder canonicalisation: B also deduplicates and may encode high-cardinality columns structurally. Our strict-threshold ablation isolates the encoding choice and shows it contributes little, and the duplicate rate is only 0.52%, bounding its contribution. Canonicalisation is therefore the dominant effect. We have not run a full factorial ablation of all four corrections.

*External validity.* Our LODO sweep covers all five devices, which resolves what would otherwise have been a serious threat: had we withheld only Modbus, we would have concluded that any unseen device breaks detection, which Table 10 shows is false. The remaining limitation is that AgriEdge contains exactly one actuation-layer device, so "the perception/actuation boundary" is inferred from a single instance of that layer. A testbed with several distinct fieldbus devices would be needed to confirm that the boundary is architectural rather than a peculiarity of this particular Modbus capture.

*Cross-dataset generalisation is out of scope, deliberately.* The natural next question is whether detectors trained on AgriEdge transfer to a different IIoT network. We do not answer it here, and the reason is worth stating plainly rather than leaving as an unexplained omission. Edge-IIoTset's features are Wireshark protocol-field names (`tcp.flags`, `mqtt.topic`); most comparable datasets—CIC-IoT-2023, TON_IoT, WUSTL-IIoT-2021—expose flow-statistical features from CICFlowMeter or Argus. Direct name-level alignment yields a near-empty intersection, so any cross-dataset number would be computed over a handful of coincidentally-shared columns and would measure the alignment more than the detector. A defensible result requires constructing a bridging feature space, which recent work [7] treats as a contribution in its own right. We judged that a second paper rather than a section of this one.

We therefore present the LODO sweep as our generalisation evidence, and we are explicit about what it does and does not establish. It shows a genuine distribution shift *within* one testbed, across an architectural layer, with effects large enough to invert model rankings. It does not establish behaviour on a different network. Readers should not read Table 10 as a substitute for cross-network validation.

*Statistical reporting.* The headline protocol comparison (Table 4) is reported over 5-fold × 3-repeat stratified cross-validation with 95% intervals. The deep baselines (Table 5), the LODO sweep, the federated runs and the edge-cost measurements are single seeded runs; their effects are large relative to the fold-level variance observed in Table 4, but we do not quantify their variance directly. Cross-validation intervals are also not strictly independent across folds within a repeat, so they should be read as descriptive of spread rather than as frequentist guarantees.

## 8. Reproducibility

All code is released under the MIT licence as the `agriedge` Python package, at `https://github.com/MostafaGalal1/agriedge`: `audit.leakage` (separation rate, single-column accuracy, NMI, provenance probe), `data.placeholders` (canonicalisation), `data.recipes` (both protocols), `data.agribench` (benchmark construction under uniform parsing), `federated` (non-IID partitioners and FedAvg), `evaluation` (metrics, repeated $k$-fold, edge cost, cross-domain alignment), and `models` (classical suite and deep baselines). Nine experiment scripts each write their own result tables.

All randomness is seeded. Every experiment reported here ran on a 12-core Apple M4 Pro with 25 GB of memory; the deep baselines used the integrated GPU through PyTorch's Metal backend, training in under a minute each. No experiment required datacentre hardware.

*Provenance of the audited copy.* Because this paper makes a claim about a distribution rather than about a model, the copy matters. Edge-IIoTset is published on IEEE DataPort, submitted by the lead author on 18 January 2022 and gated behind a subscription [16]. The DataPort record itself, however, instructs readers to retrieve the data from a Kaggle repository under that same author's account, and gives the retrieval path verbatim as `Edge-IIoTset dataset/ Selected dataset for ML and DL/DNN-EdgeIIoT-dataset.csv`. The free channel is therefore author-sanctioned rather than a third-party re-upload, it serves the same release as the subscription channel, and—being free—it is what the community in practice trains on. The copy audited here was obtained through it, at exactly that path.

Three further observations argue that the artifact is not an accident of repackaging. The copy's `Readme.txt`, documentation PDF and CSV files all carry the authors' 18 March 2022 timestamps, across ten normal-traffic device directories and fourteen attack captures, and its structure matches the description in the dataset paper: 61 protocol features alongside `Attack_label` and `Attack_type`. The artifact is present in the authors' own per-device raw captures, not only in the merged curated subsets—the normal-traffic captures write the placeholder as `0` and the attack captures write it as `0.0`—and a mirror that re-archives a download does not regenerate two dozen per-device CSVs. And it is visible in the raw CSV text, independent of any parsing library.

We nonetheless encourage readers to verify against their own copy. The repository includes a dependency-free script that does so in one command, reading every field as text so that no type inference can mask the difference:

```
python scripts/verify_placeholder_split.py \
       ML-EdgeIIoT-dataset.csv
```

The audit reduces to a single call for readers who wish to check their own pipelines:

```
from agriedge.audit.leakage import (
    audit_columns,
    summarize,
)

reports = audit_columns(
    frame,
    categorical_columns,
    label_column="Attack_label",
)
print(summarize(reports))
```

## 9. Conclusion

Edge-IIoTset carries a serialisation artifact that makes file provenance linearly separable from the label. The preprocessing recipe distributed with the dataset converts a placeholder-spelling difference between the normal-traffic and attack-traffic build branches into features that recover the label exactly. Four of the seven columns the recipe names separate every row of both curated subsets on their own—157,800 rows in one, 2,219,201 in the other—and under that recipe five of six standard classifiers attain exactly 1.0000 accuracy in every one of fifteen cross-validation folds and the sixth attains 0.99998, while a 1D-CNN and a deep MLP reach 1.0000 on a single stratified split. The leak is not a property of one-hot encoding: label, ordinal and frequency encoding transmit it identically.

We stop short of claiming that any particular published number is an artifact, since preprocessing is rarely reported in sufficient detail to determine it from outside. What we can say is stronger than a suspicion and weaker than an indictment: the recipe the dataset's own documentation prescribes yields perfect or all-but-perfect classification for every model we tried; the encodings researchers substitute for it do the same; and the affected columns are the ones an IIoT practitioner would naturally reach for. Any result on this dataset whose preprocessing touched those columns needs re-checking before it can be read as detection performance.

Correcting the artifact costs the strongest model about five points of macro-F1 and costs the weakest nearly a third of its score. Rebuilding the benchmark from raw captures under uniform parsing eliminates the artifact by construction, restores the Modbus traffic and per-device attribution that agricultural research requires, and yields a benchmark on which no column separates the classes above 0.0288.

On that corrected benchmark, a leave-one-device-out sweep locates where detection actually stops working. Transfer between perception sensors is nearly free. Transfer across the perception/actuation layer is not: withholding the Modbus gateway drops random forest from 0.9988 to 0.5083 balanced accuracy and logistic regression to 0.4877, straddling the trivial baseline, and inverts the model ranking so that optimising average performance selects the model least able to survive the shift. Combined with a 342× spread in inference latency across models of indistinguishable random-split accuracy, the numbers the field currently optimises say very little about deployment.

Our federated results sharpen the same point from a different direction. Non-IID partitioning—the difficulty the federated IIoT literature most often sets out to solve—costs at most 0.0037 macro-F1 on this benchmark, while the uplink required to train even a 28,450-parameter model over LoRaWAN costs 4.6 hours of continuous transmission. The obstacle to federated intrusion detection on a real farm is bandwidth, not statistical heterogeneity.

The corrective is inexpensive: canonicalise placeholders before encoding, deduplicate before splitting, evaluate under held-out-domain protocols, include a weak model as a leakage canary, and report deployment cost alongside accuracy. We release tooling for all five.

### *9.1. Future work*

Three directions follow directly. **Cross-network transfer**: establishing whether detectors trained on AgriEdge survive a genuinely different network requires a bridging feature space between packet-field and flow-statistical representations, which is the principal open problem for this line of work. **A second actuation-layer device**: our central generalisation finding rests on a single fieldbus device; a testbed carrying several distinct actuation protocols (Modbus alongside DNP3, BACnet, or OPC-UA) would determine whether the boundary is a property of the layer or of this particular capture. **Auditing other security datasets**: the failure mode requires only that a dataset be assembled by concatenating separately-parsed sources whose separation correlates with the label, which describes a large fraction of intrusion-detection corpora. Applying the audit released here across the standard benchmark suite would establish whether Edge-IIoTset is exceptional or representative. We suspect the latter.

## Data availability

Edge-IIoTset is publicly available and free for academic use per its own licence [1]. The `agriedge` package, the AgriEdge benchmark construction pipeline, and all experiment scripts and result tables are released under the MIT licence at `https://github.com/MostafaGalal1/agriedge` and archived under the persistent identifier `https://doi.org/10.5281/zenodo.21941210`, which resolves to the most recent release. The constructed benchmark is not carried in the repository, being 38 MB; it is published as a separate archived dataset at `https://doi.org/10.5281/zenodo.21941319`, so that it can be obtained directly without first acquiring the 10 GB source distribution. It is equally regenerable by running `experiments/03_build_agribench.py` against a local copy of Edge-IIoTset.

## Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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